Rebecca Woody

Ph.D. Candidate, Astronomy & Astrophysics

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Research Interests

Stellar evolution and modeling, stellar ages, isochrone fitting, asteroseismology, exoplanet ages, metal-poor stars, Galactic archaeology, convection and mixing length theory, low mass stars, stellar spectroscopy, globular clusters

Education

PhD **Harvard University**, Astronomy & Astrophysics — Cambridge, MA
- Advisor: Charlie Conroy — Aug 2021 – May 2027 (exp.)

BS, BA **Johns Hopkins University**, Physics, Mathematics — Baltimore, MD
- Summa cum laude — Aug 2017 – May 2021
- Advisor: Kevin C. Schlaufman, Keith Hawkins

Professional Appointments

- Graduate Research Assistant, Center for Astrophysics | Harvard & Smithsonian — 2021 – Present
- NSF REU Research Assistant, University of Texas at Austin — 2019
- Undergraduate Research Assistant, Johns Hopkins University — 2018 – 2021

Teaching

- Harvard University — Lab Teaching Fellow (Spring 2024): ASTRON 16, Stellar and Planetary Astronomy
- Harvard University — Teaching Fellow (Spring 2022): ASTRON 100, Methods of Observational Astronomy

Honors and Awards

- Donald E. Kerr Memorial Award, Johns Hopkins University Department of Physics & Astronomy — 2021
- Goldwater Scholar — 2020

Talks

- Astro Coffee, Institute for Astronomy (IfA), University of Hawai‘i — November 2026
- Student Faculty Forum (StuFF), Center for Astrophysics | Harvard & Smithsonian — October 2026
- Exoplanets & Stars Seminar, Yale University — September 2026
- Yale/CeNAM Nuclear Asteroseismology Workshop, Yale University — July 2026
- MMT Science Symposium, University of Arizona — April 2026
- MPIA Stars Meeting — July 2023
- ITC Luncheon Talk, Center for Astrophysics | Harvard & Smithsonian — April 2023
- Houston Astronomical Society — August 2018
- Research Symposium, Maryland Space Grant Consortium — June 2018

Posters

- Searching for r-process Enhanced Metal-poor Stars in the Milky Way Halo, AAS 235 — January 2020

Workshops & Conferences

- Yale/CeNAM Nuclear Asteroseismology Workshop, Yale University, New Haven, CT — July 2026
- MESA Summer School, KU Leuven, Belgium — July 2025
- Binary Stars in the Space Era, Keele University, UK — July 2025
- SDSS-V Collaboration Meeting, Flatiron Institute, NYC, NY — August 2023
- 235th AAS Meeting, Honolulu, HI — January 2020

Observatory Allocations

- Magellan Observatory, MIKE, 9 nights — 2023

First-Author Publications

Classical Cases of Non-solar Mixing Length in Low Mass Stars — 2026
Woody, Rebecca, Conroy, C., Cargile, P. A., Dotter, A.
arxiv.org/abs/2609.13384 (submitted to ApJ; arXiv:2609.13384)

The Rapid Formation of the Metal-poor Milky Way — 2025
Woody, Rebecca, Conroy, C., Cargile, P., et al.
ui.adsabs.harvard.edu/abs/2025ApJ…978..152W/abstract (ApJ, 978, 152)

The Age–Metallicity–Specific Orbital Energy Relation for the Milky Way's Globular Cluster System Confirms the Importance of Accretion for Its Formation — 2021
Woody, Rebecca, Schlaufman, K. C.
ui.adsabs.harvard.edu/abs/2021AJ….162…42W/abstract (AJ, 162, 42)

Co-Author Publications

An Ancient Brown Dwarf Transiting a Metal-poor Thick-disk Star — 2026
Adams Redai, J., Chandra, V., Yee, S. W., …, ***Woody, Rebecca***, et al.
ui.adsabs.harvard.edu/abs/2026ApJ..1002L..41A/abstract (ApJL, 1002, L41)

All-sky Kinematics of the Distant Halo: The Reflex Response to the LMC — 2025
Chandra, V., Naidu, R. P., Conroy, C., …, ***Woody, Rebecca***, et al.
ui.adsabs.harvard.edu/abs/2025ApJ…988..156C/abstract (ApJ, 988, 156)

A Detailed Chemical Study of the Extreme Velocity Stars in the Galaxy — 2024
Nelson, T., Hawkins, K., Reggiani, H., …, ***Woody, Rebecca***
ui.adsabs.harvard.edu/abs/2024MNRAS.532.2875N/abstract (MNRAS, 532, 2875)

Distant Echoes of the Milky Way's Last Major Merger — 2023
Chandra, V., Naidu, R. P., Conroy, C., …, ***Woody, Rebecca***, et al.
ui.adsabs.harvard.edu/abs/2023ApJ…951...26C/abstract (ApJ, 951, 26)

Dwarf Galaxy Archaeology from Chemical Abundances and Star-formation Histories — 2023
Johnson, J. W., Conroy, C., Johnson, B. D., …, ***Woody, Rebecca***, et al.
ui.adsabs.harvard.edu/abs/2023MNRAS.526.5084J/abstract (MNRAS, 526, 5084)

A Ghost in Boötes: The Least-Luminous Disrupted Dwarf Galaxy — 2022
Chandra, V., Conroy, C., Caldwell, N., …, ***Woody, Rebecca***
ui.adsabs.harvard.edu/abs/2022ApJ…940..127C/abstract (ApJ, 940, 127)

The Stellar Halo of the Galaxy is Tilted and Doubly Broken 2022
Han, J. J., Conroy, C., Johnson, B. D., ..., ***Woody, Rebecca***, et al.
ui.adsabs.harvard.edu/abs/2022AJ....164..249H/abstract (AJ, 164, 249)

A Tilt in the Dark Matter Halo of the Galaxy 2022
Han, J. J., Naidu, R. P., Conroy, C., ..., ***Woody, Rebecca***
ui.adsabs.harvard.edu/abs/2022ApJ...934...14H/abstract (ApJ, 934, 14)

Birth of the Galactic Disk Revealed by the H3 Survey 2022
Conroy, C., Weinberg, D. H., Naidu, R. P., ..., ***Woody, Rebecca***, et al.
ui.adsabs.harvard.edu/abs/2022arXiv220402989C/abstract (submitted to OJAp; arXiv:2204.02989)

Live Fast, Die α-Enhanced: The Mass–Metallicity–α Relation of the Milky Way's Disrupted Dwarf Galaxies 2022
Naidu, R. P., Conroy, C., Bonaca, A., ..., ***Woody, Rebecca***, et al.
ui.adsabs.harvard.edu/abs/2022arXiv220409057N/abstract (submitted to ApJ; arXiv:2204.09057)